

Task 5 – Model Analysis and Documentation

Dataset

- **Dataset:** CIFAR-10
- **Total Images:** 60,000
- **Training:** 50,000
- **Testing:** 10,000
- **Classes:** 10
- **Image Size:** 32×32×3
- **Classes:** Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship, Truck

Training Details

CNN

- **Optimizer:** Adam
- **Loss:** Sparse Categorical Crossentropy
- **Epochs:** 10
- **Batch Size:** 64
- **Input Size:** 32×32×3

MobileNetV2

- **Pretrained ImageNet weights**
- **Base model frozen**
- **Input Size:** 96×96×3
- **Optimizer:** Adam
- **Loss:** Sparse Categorical Crossentropy
- **Epochs:** 10
- **Batch Size:** 64

Model Comparison

Model	Accuracy
CNN	70%

Model	Accuracy
MobileNetV2	81%

Performance Analysis

MobileNetV2 achieved **81% accuracy**, compared with **70%** for CNN.

Advantages & Limitations

CNN

- Simple and customizable
- Learns features directly
- Requires more training

Transfer Learning

- Faster training
- Uses pretrained features
- Better performance
- Depends on a pretrained model

Challenges

- Image preprocessing and resizing
- Training time
- Improving accuracy
- Classifying similar images

Applications

- Image recognition
- Object classification
- Autonomous systems
- Medical image analysis

Future Improvements

- Data augmentation
- Fine-tuning MobileNetV2
- Hyperparameter tuning
- Testing other pretrained models